12-GAUGE SIDE-BY-SIDE DOUBLE-BARREL SHOTGUNS

HISTORICAL CONTEXT

During the 1920s, Sears, Roebuck & Co. contracted America's foremost arms makers?primarily the J. Stevens Arms Company (Chicopee Falls, MA) and the Crescent Fire Arms Co. (Norwich, CT)?to manufacture durable boxlock and sidelock side-by-side shotguns marketed under the 'Ranger' and store house brands.

PURPOSE OF THIS MANUAL

This technical dossier provides master gunsmith specifications, headspace gauging limits, hinge-pin fitting procedures, sear overlap geometry, and the mandatory pre-1927 chamber length safety protocol.

Anson & Deeley Boxlock Action Principles

ACTION LOCKUP SPECIFICATIONS

* Hinge Pin Diameter: 0.375" nominal hardened tool steel.
* Hook Bearing Area: Minimum 80% uniform bearing contact across hinge pin diameter.
* Off-Face Clearance: 0.000" (Zero allowable clearance between breech face and barrel breech).
* Top Snap Lever Angle: Must rest at 4:30 to 5:00 o'clock position when closed.
* Underbolt Bite: Full engagement into rear barrel lump notch.

BARREL ASSEMBLY & CHOKE BORING

* Construction: Fluid Compressed Steel or High-Pressure Decarb Steel; concave matted rib.
* Standard Lengths: 28", 30", and 32" options.
* Choke Constrictions: Right Barrel = Modified (~0.020" constriction); Left Barrel = Full (~0.035" constriction).
* Bore Diameter (Nominal 12G): 0.729" to 0.732".

Chamber Length Inspection Protocol

·· MANDATORY SAFETY NOTICE: SHORT CHAMBERS

Prior to 1927, many American side-by-side shotguns were chambered for 2-5/8" or 2-1/2" paper hulls using roll crimps. Modern plastic hulls measure 2-3/4" AFTER firing.

If a modern star-crimped 2-3/4" high-velocity cartridge is fired in a 2-5/8" chamber, the opened plastic crimp petals protrude directly into the forcing cone. This creates a severe bore restriction, spiking peak breech pressure from standard 11,500 psi to over 15,000?18,000 psi.

RESULT: Blown action off-face, cracked breech frame, or catastrophic burst barrel.

GUNSMITH INSPECTION PROCEDURE

1. Clean both chambers completely free of powder and lead residue.
2. Insert a Brownells 12-Gauge Chamber Length Gauge until its rim shoulder seats on the barrel rim cut.
3. Verify that the chamber length shoulder measures a full 2-3/4" (70mm) to the start of the forcing cone.
4. If short, either lengthen the chamber and forcing cone using a 12G chambering reamer with 1-1/2" gradual forcing cone, or strictly label the firearm: 'LOW-PRESSURE SHORT HULLS ONLY (Max 7,500 psi)'.

Tolerances & Bench Remedies

12-GAUGE RIM HEADSPACE TOLERANCES

* GO Gauge (Minimum Rim Recess): 0.054" ? Barrels must lock shut without forcing.
* NO-GO Gauge (Max New Rim Recess): 0.064" ? Action must NOT fully lock shut.
* FIELD Gauge (Service Safety Maximum): 0.070" ? If action locks on 0.070", UNSAFE TO FIRE.

REPAIRING AN 'OFF-FACE' SHOTGUN

Symptom: With forend removed, barrels shake forward/rearward or rattle against frame.

Step 1: Check frame hinge pin for wear or indentation using micrometers.
Step 2: If pin is worn, drive out frame pin and ream hole to accept a +0.005" or +0.010" precision oversize hinge pin.
Step 3: If barrel hook is indented, weld hook face using micro-TIG (ER70S-6 wire) or silver-solder a precision steel shim.
Smoke-test or Prussian blue against hinge pin and hand-file with riffler files until 85%+ uniform bearing contact is achieved.

Armorer Tuning & Safety Engagement

TRIGGER PULL WEIGHTS & SEAR GEOMETRY

* Front Trigger (Right Barrel): 4.0 to 4.5 lbs.
* Rear Trigger (Left Barrel): 4.5 to 5.0 lbs.
* Firing Pin Protrusion: 0.055" to 0.065"; radius tips to smooth hemisphere.
* Sear Engagement Overlap: 0.030" minimum full square overlap.

PREVENTING 'DOUBLING' (BOTH BARRELS FIRING TOGETHER)

Cause: When front trigger is pulled, heavy recoil dislodges the left hammer sear.

Bench Remedy:
1. Detail strip action and clean hardened animal-fat grease from sear pivots.
2. Stone hammer notch and sear face strictly square with hard Arkansas oilstone at 90°.
3. Verify left sear return spring free length and tension; replace fatigued springs.
4. Test by cocking both hammers and striking rubber pad on bench with sharp mallet blow; hammers must remain locked.

Sears Store Brand Gunsmith Identification Directory

HOW TO IDENTIFY SEARS SHOTGUN CONTRACTORS

Sears stamped store brand firearms with a 3-digit manufacturer prefix preceding the model number:

? Code 101.xxxx -> J. Stevens Arms Co. / Savage Arms (Stevens 311, 5100, Springfield 5100)
? Code 105.xxxx -> Crescent Fire Arms Co. (Empire Ejector, Peerless, Triumph)
? Code 103.xxxx -> Marlin Firearms Co.
? Code 102.xxxx -> Savage Arms Corporation
? Meriden Fire Arms -> Wholly owned by Sears in Meriden, CT (1905?1918; A.J. Aubrey)
? Eastern Arms Co. -> Commercial contract brand for Sears distributed hardware doubles.

SUMMARY OF MAINTENANCE PROTOCOLS

1. Never fire modern 3" Magnum shells in 2-3/4" or vintage 2-5/8" chambers.
2. Always grease the hinge pin and forend iron knuckle with heavy moly or lithium grease.
3. Keep action lockup clean; avoid slamming barrels shut (ease action closed and release top lever).
4. Use snap caps when dry firing to protect vintage striker noses from peening.